

BSAA

Agentic Bank Statement Analysis Platform — Indian Market

System Tour · Sprint 4

The Problem

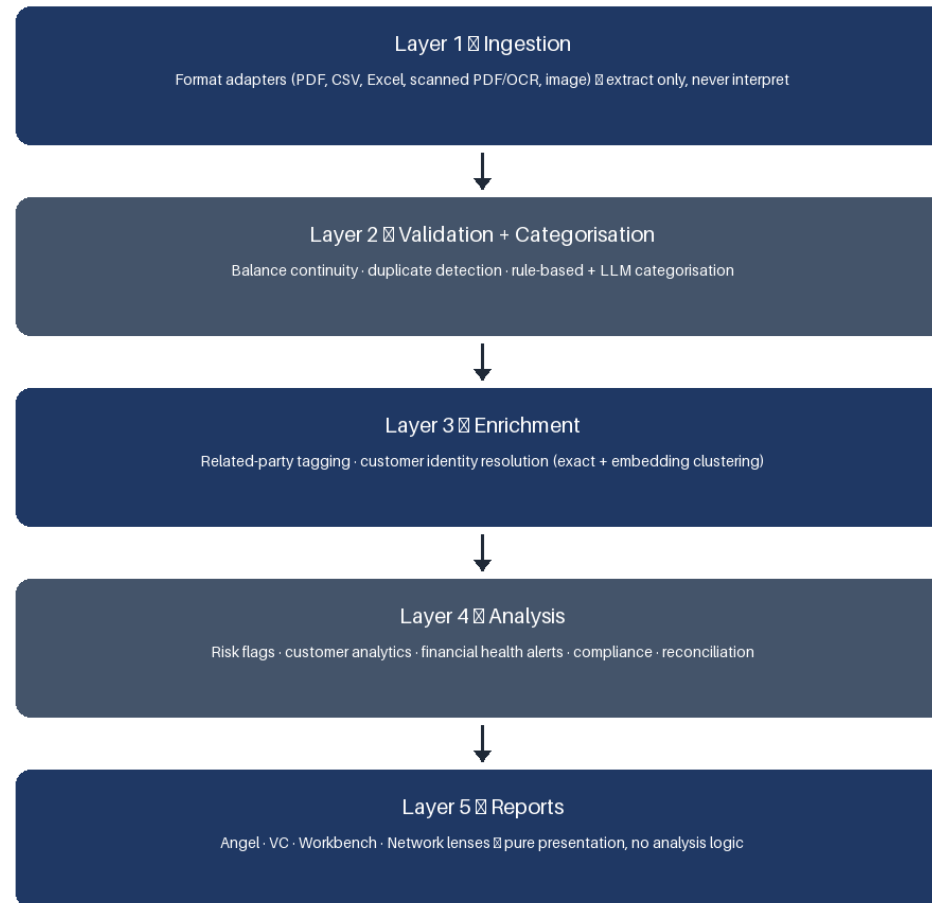
Manual Indian bank statement review for investor due diligence is slow, inconsistent, and doesn't scale — and different banks (HDFC, ICICI, SBI, Axis, Kotak) all use different PDF layouts.

One Engine, Many Lenses

A single analysis pipeline produces every finding once. Four lightweight presentation templates ("lenses") tailor that same output for angel investors, angel networks, VCs, and in-house finance teams — adding a new user type means a new template, not a new engine.

Architecture — 5-Layer Pipeline

BSAA 5-Layer Pipeline



Layer 1 — Ingestion

- Format adapters: digital PDF, CSV, Excel, scanned PDF (OCR), image
- Adapters extract only — they never decide what a column means
- HDFC + ICICI supported today; profile-based, so new banks are new profiles

Layer 2 — Validation + Categorisation

- Balance continuity is the primary defence against silent data corruption
- Duplicate detection, date-order and gap checks
- Rule-based categorisation covers ~98% of cases; Claude Haiku LLM fills the rest

Layer 3 — Enrichment

- Related-party tagger: exact match + LLM fuzzy match against affiliate list
- Customer identity resolver: exact-match clustering + sentence-embedding merge
- Every REVENUE transaction ends up with a stable customer_id

Layer 4 — Analysis

- Six independent risk detectors + composite 0-100 score
- Customer analytics: NRR, churn, cohort retention, concentration trajectory
- Compliance: India jurisdiction module (§269ST, GST, TDS, PMLA, related-party limits)
- Reconciliation: bank-statement reality vs. declared pitch-deck claims

Reports — Angel & VC Lens

Angel Lens

1-page PDF: verdict, KPIs, runway gauge, health signals — the quick read for angel investors.

VC Lens

6-sheet XLSX + 3-page PDF: financial health, risk, customer analytics, transactions, related parties, composite risk score.

Reports — Workbench & Network Lens

Workbench Lens

10-sheet XLSX + 4-page PDF: the deepest view, with fund-specific custom detectors via `workbench_config.py`.

Network Lens

Cross-company comparison workbook for angel investor networks — sortable table, risk distribution, cohort dashboard.

Validation Summary

Metric	Value
Gold-set statements	15
Transaction extraction accuracy	100.0%
Categorisation accuracy	99.5%
Sprint 1 targets met	Yes

759 automated tests across the full pipeline (schema, adapters, normaliser, validator, categoriser, risk, customer analytics, compliance, reconciliation, batch processing, and all four report lenses).

What's Next

- Account Aggregator (Sahamati) sandbox adapter — proof that the pattern extends to live consent-based data, pending sandbox credentials
- Production AA integration beyond the sandbox proof-of-concept
- Additional jurisdiction modules for markets beyond India
- Fund-specific onboarding: custom detectors and lens branding per `workbench_config.py`